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What Makes a Good Image? Airbnb Demand Analytics Leveraging Interpretable Image Features

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Abstract. We study how Airbnb property demand changed after the acquisition of *verified* images (taken by Airbnb's photographers) and explore what makes a good image for an Airbnb property. Using deep learning and difference-in-difference analyses on an Airbnb panel data set spanning 7,423 properties over 16 months, we find that properties with verified images had 8.98% higher occupancy than properties without verified images (images taken by the host). To explore what constitutes a good image for an Airbnb property, we quantify 12 human-interpretable image attributes that pertain to three artistic aspects—composition, color, and the figure-ground relationship—and we find systematic differences between the verified and unverified images. We also predict the relationship between each of the 12 attributes and property demand, and we find that most of the correlations are significant and in the theorized direction. Our results provide actionable insights for both Airbnb photographers and amateur host photographers who wish to optimize their images. Our findings contribute to and bridge the literature on photography and marketing (e.g., staging), which often either ignores the demand side (photography) or does not systematically characterize the images (marketing).

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1. Introduction

The global sharing economy market has been rapidly expanding in recent years and is projected to generate roughly $335 billion by 2025 (PwC 2015). On Airbnb, travelers can find lodging in home environments rather than hotels, and hosts can generate rental income from spare rooms or properties. As the world's largest home-sharing platform, Airbnb recently was valued 20% higher than Marriott and hosted 25% more guests per night than Hilton Worldwide (Winkler and Macmillan 2015).

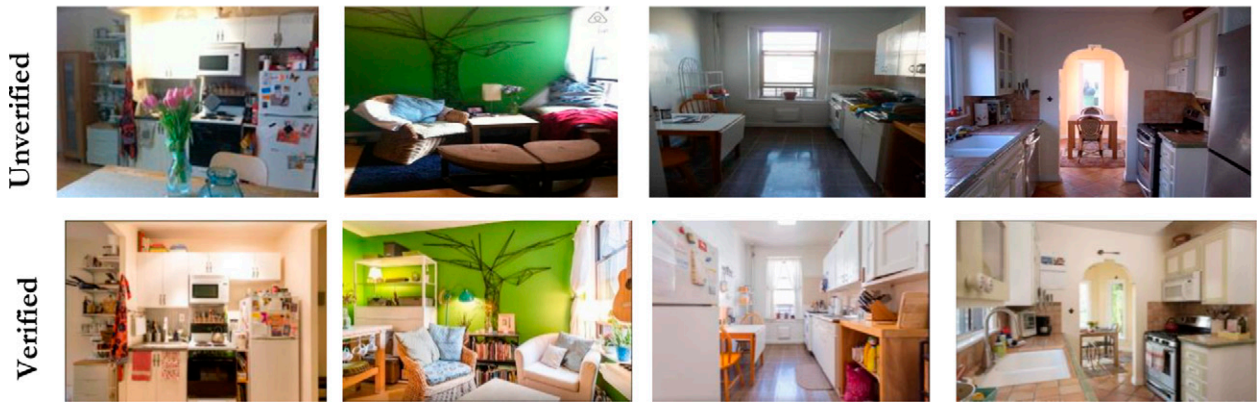
Despite its success, Airbnb faces a significant problem in resolving the uncertainty that consumers face when evaluating property quality. The inefficiency of transferring information required by prospective guests creates transactional friction and leads to the loss of users. Reports show that quality uncertainty leads many potential consumers to choose trusted hotel brands over Airbnb (PwC 2015, Ufford 2015).

Airbnb attempts to alleviate quality uncertainty with features such as customer reviews, host verification, property descriptions, and property images. In particular, property images provide visual information and reduce uncertainty about experiential aspects (e.g., cleanliness and mood) of units in ways that written reviews and descriptions cannot. Although hotel images are taken by professional photographers, most Airbnb property images are taken by hosts—usually amateur photographers. Therein lies a source of inefficiency of information transfer that causes uncertainty for potential guests. Hosts often lament the poor quality of their own amateur photos and fear that their properties appear small.

To address this concern, in 2011, Airbnb launched a "photography program" that gives interested hosts free access to local professional photographers who are commissioned by the company to take photos of the host's property. A "verified" mark appears on images shot and uploaded by Airbnb's professional photographers. Figure 1 juxtaposes the original amateur photographs with Airbnb's professional photographs of the same room to illustrate the improvement.

It remains unclear, however, whether the professional photography program has been beneficial for Airbnb and its hosts. In fact, the photography program has raised much controversy among Airbnb hosts and guests. It seems plausible that verified photos could attract more guests, but they also could

Figure 1. (Color online) Comparison of Unverified and Verified Photos



oversell or misrepresent the property, leading to a negative impact on property demand—a fear voiced by some Airbnb hosts on the host forum.[1] The Airbnb photography program raises a series of questions. (1) Do properties with verified photos experience higher demand? (2) If so, what systematic differences between verified and unverified photos might explain the higher demand? (3) Finally, what constitutes a good image for an Airbnb property?

To answer these questions, we collected panel data from 13,000 Airbnb listings with over 510,000 property images in seven major U.S. cities from January 2016 through April 2017. The data set contains rich information about each property's monthly reservations, photos, price, and other details about the property and host. The data set captures variation in property images across both units and time, so we can observe some of the properties transition from using unverified photos to verified photos.

We use Amazon Mechanical Turk (AMT), a crowdsourcing platform for on-demand human tasks, to classify a random set of pictures into the binary categories of high and low quality. We use the manually classified training set to develop a scalable model that takes advantage of computer vision and deep learning. Specifically, taking pixel-level information from the images as input, we apply a convolutional neural network (CNN) to classify the aesthetic quality of each image in the training sample. The CNN model is optimized to extract a hierarchical set of features from images and learn the relationship between the set of features and the image's label (in this case, "high quality" or "low quality"). Then, we use our trained CNN image quality classifier to label the property images in the Airbnb data set.

We employ a difference-in-difference (DiD) analysis to compare the demand for properties with verified photos and with unverified photos. Hosts self-select into the Airbnb photography program, so we address endogeneity concerns through propensity score weighting (PSW). The PSW approach assumes that selection into the treatment group is captured by observed variables, and the algorithm matches treated and untreated units using estimated treatment probabilities. The PSW approach mitigates the self-selection problem only so far as the factors that affect both the treatment probability and property demand are captured in observed variables. Thus, we conduct a Rosenbaum bounds analysis to test the robustness of our results to hypothetical unobservables. The results of the Rosenbaum analysis are within the range reported in prior literature, and a series of other tests suggests similarly robust results. Nevertheless, we cannot fully rule out the possibility that selection is driven by unobserved variables in the absence of a random experiment, and we caution readers to interpret our results with this caveat in mind.

We find that the occupancy rate is 8.985% higher for properties with verified photos than for properties with unverified photos. The treatment coefficient is positive and significant even after controlling for other sources of information, such as guest reviews. The estimated treatment coefficient decreases dramatically after we incorporate photo characteristics into the demand model. Particularly, the coefficient decreases by 41.0% when we control for image quality, suggesting that a significant portion of the coefficient is explained by the higher quality of the verified images.

One of our key objectives is to determine what makes a good image for an Airbnb property. Our CNN model is highly accurate at predicting image quality, but the CNN-extracted features are uninterpretable. To provide better guidance for managers, we use the photography literature to identify 12 human-interpretable image attributes that are relevant to image quality in the real estate context. We theorize the relationship between each of the 12 image attributes and property demand. The 12 attributes fall under three key artistic aspects: composition, color, and the figure-ground (FG) relationship. Composition is the arrangement of visual elements in the photograph; ideally, the composition leads the viewer's eyes to the center of focus (Freeman 2007). We capture composition with four attributes: diagonal dominance, the rule of thirds (ROT), visual balance of color, and visual balance of intensity. Color can affect the viewer's emotional arousal. The marketing literature has studied the impact of color on consumer behavior particularly in the context of web design, product packaging design, and advertisement design (Gorn et al. 1997, 2004; Miller and Kahn 2005). We include five aspects related to color: warm hue, saturation, brightness, contrast of brightness, and image clarity. The principle of the figure-ground relationship is one of the most basic laws of perception and is used extensively by expert photographers to plan their photographs. In visual art, the *figure* refers to the key region (i.e., foreground), and the *ground* refers to the background; photographs in which the figure is inseparable from the ground do not retain the viewer's attention. We include three attributes: area difference, texture difference, and color difference between the figure and ground.

We use computer vision methods to score the images on the 12 image attributes, and we find systematic differences between the *verified* photos and both the unverified low-quality and unverified high-quality photos. The verified photos dominate the unverified high-quality photos on all attributes except for the rule of thirds and saturation. The verified photos score higher than the unverified high-quality photos across the attributes for which the theorized effect of the attribute is positive; however, the verified photos score lower than the unverified high-quality photos when the theorized effect of the attribute is negative.

We explore how the 12 image attributes might be related to property demand, although we cannot identify causal effects of these 12 image attributes on property demand given the observational nature of our data. That is, the image attributes change when a property acquires verified photos, but the choice to do so is endogenous, making any changes in the image attributes also endogenous. As a result, our analysis of the relationship between the image attributes and property demand should be viewed as exploratory. After controlling for the 12 attributes, the estimated treatment coefficient becomes statistically insignificant. The results suggest that the positive coefficient of verified photo acquisition may derive from the 12 image attributes, which together capture both vertically differentiated quality and horizontally differentiated taste. Airbnb professional photographers seem to be better than host photographers at capturing the attributes that matter for Airbnb property demand.

Of the 12 image attributes, the visual balance of color is most strongly related to property demand, followed by image clarity and the contrast of brightness. The visual balance of color refers to color symmetry, which can be affected by both the property itself and the position from which the image is captured. Image clarity refers to the extent to which the image conveys visual information. The unverified low-quality images scored poorly on image clarity; the verified photos scored almost twice as high. Even without employing a professional photographer, hosts can improve image clarity through the effective use of lighting and access to a good camera. Finally, the contrast of brightness captures the difference in illumination between the brightest and dimmest points in the image; a low contrast of brightness indicates that illumination is relatively even across the image. The verified photos have a significantly lower contrast of brightness than unverified high-quality images. Interestingly, several hosts on the Airbnb community forums complained that the contrast of brightness is so low in the verified photos that they appear washed out, but we find the predicted negative relationship between the contrast of brightness and property demand. In other words, consumers seem to prefer the low contrast of brightness that appears in verified photos.

This study makes several contributions. The ability to capture and analyze unstructured data, particularly images, is a nascent field with few papers (e.g., Wang et al. 2018, Malik and Singh 2019, Malik et al. 2019, Netzer et al. 2019, Liu et al. 2020). Ours is among the first to examine how Airbnb property demand changed with the adoption of the Airbnb photography program. From a managerial perspective, our results suggest that the program is successful for hosts who choose to take advantage of it—properties with verified images achieved an 8.98% higher occupancy rate than properties without verified images. However, more than half of the Airbnb properties use low-quality images. Our findings may motivate hosts to obtain high-quality images of their properties.

Although it seems intuitive that the demand should be higher for properties with good images, the components of a "good image" are less obvious. A key practical finding of our paper is the identification of 12 interpretable image attributes that significantly correlate with property demand. Our investigation has parallels with marketing studies on color as a key image attribute that affects consumers' perceptions and product demand. Beyond color, we consider composition and the figure-ground relationship, and we find that even the high-quality unverified images are systematically inferior to the verified images on most of the 12 attributes. Our findings may be valuable to photographers who are attempting to optimize the appeal of their images for Airbnb consumers.

Moreover, insights from our paper can guide image content engineering efforts in short-term lodging contexts beyond Airbnb (e.g., hotels) and real estate markets. Our approach efficiently computes an extensive set of image attributes (at ~1.06 seconds per image) on a 2-Intel-Haswell (E5-2695 v3) central processing unit, so it can be implemented as a scalable real-time model. Our application of an existing method for image quality classification and feature extraction can be adopted by photographers of firms and hosts to check their image quality and identify the shortcomings of their photographs. Such a practice ultimately could improve the efficiency of image-based information transfer and may increase the appeal of their properties.

Prior studies in the marketing literature have considered the effects of images and visual elements, but our paper differs from existing papers in three ways. First, the extant studies focus on the viewer's emotional arousal and are restricted to either isolated image features (e.g., color; Gorn et al. 1997, 2004) or high-level image content or style (e.g., whether an image is "art"; see Hagtvedt and Patrick 2008). By contrast, we use the photography literature to identify 12 image attributes on which any photograph can be evaluated, and we explore their relationships with property demand. Second, the relevant literature focuses primarily on high-quality images (Bertrand et al. 2010), but e-commerce (including the sharing economy) often uses low-quality, user-generated product images. Finally, the evolving stream of marketing literature on the impact of images on consumers' product perceptions focuses on the mere presence or absence of an image rather than specific image attributes (e.g., Mitchell and Olsen 1981; Meyers-Levy and Peracchio 1992; Peracchio and Meyers-Levy 1994, 2005; Scott 1994; Valdez and Mehrabian 1994; Gorn et al. 1997; Larsen et al. 2004; Miller and Kahn 2005).

2. Empirical Framework

2.1. Data Description

We randomly selected more than 13,000 Airbnb property listings from seven U.S. cities (Austin, Boston, Los Angeles, New York, San Diego, San Francisco, and Seattle), and we collected data on the listings from January 2016 to April 2017. We obtained information about each property host from that host's public profile on Airbnb.com. Each profile specifies the date on which the host joined Airbnb and whether the host had a verified Airbnb account at the time of the analysis. For each property, we obtained information about static characteristics: location (city and zip code), property type (e.g., house, apartment), property size (number of beds), amenities (e.g., pool, air conditioning, proximity to a beach), and capacity (maximum guests). We also obtained information about dynamic characteristics: property bookings, nightly prices, guests' reviews, property photos, and whether the photos were verified. In the online appendix (Section VI), we provide detailed methods for data collection and matched sample construction. In the next section, we describe the measures of our key variables and summarize their measurement statistics.

2.2. Definitions and Measures of Key Variables

2.2.1. Treatment Group, Untreated Group, and Treatment Status. The panel data cover 16 one-month periods from January 2016 through April 2017. The "treatment" is the adoption of verified photos; therefore, a property is "treated" if it used verified property photos during the observation period. The sample for our main analysis consists of 7,423 unique properties that did not have verified photos in January 2016;[2] 212 had verified photos by the end of April 2017 (the treatment group), and the remaining 7,211 properties did not (the untreated group). We capture the treatment with three indicator variables. $TREAT_i$ equals one (zero) if property i belongs to the treatment group (untreated group). $AFTER_{it}$ equals one (zero) if period t is after (before) the period when property i was first observed to have verified photos. For example, if a property acquired verified property photos in March 2016, then $AFTER_{it}$ equals zero for the periods of January and February 2016 and equals one for the periods of March 2016 onward. Hence, the treatment status indicator $TREATIND_{it} = TREAT_i \times AFTER_{it}$ equals one if i was treated in period t and equals zero otherwise.

2.2.2. Property Demand. We purchased listing-level booking data from a company that specializes in collecting Airbnb property demand data. The booking data include the number of days in a month in which the property was open (i.e., available to be booked), blocked (i.e., marked as unavailable by the host), and booked (i.e., by a guest). For property i in period t, we operationalize property demand as the occupancy rate: that is, the fraction of the open days that were booked, scaled by 100. For example, if a property in March was open for 24 days and booked for 6 days, then its demand for that month was $DEMAND_{it} = (6/24) \times 100 = 25.00$.

2.2.3. Property Price. The property price for property i in period t, $NIGHTLY\_RATE_{it}$, refers to the average nightly price over the days in period (Zhang et al. 2019). Property price is endogenous because it correlates with random demand shocks in the current period, which also affect property demand. To address the endogeneity concern, we use a set of instrument variables (IVs) for price. Following the extant literature, we include the characteristics of competing properties (Berry et al. 1995, Nevo 2000). The logic is that the characteristics of competing products are unlikely to be correlated with unobserved shocks in the demand for the focal property. However, the proximity of the characteristics of a property and its competitors influences the competition and as a result, the property markup and price.[3] In addition, we collect cost-related variables: that is, the factors that enter the cost/supply side but not the demand side. We use the local (zip code-level) residential utility fee obtained from OpenEI and local rental information collected from Zillow.[4] These factors affect the host's expenses and thus, provide an indirect measure of cost, but they are unlikely to be correlated with demand in the short-term lodging market.

2.2.4. Property Photos. The property photos are the set of photos posted on the property web page in the given period. Three variables characterize the property photos: photo quantity, photo quality, and the distribution of photographed room types. $IMAGE\_COUNT_{it}$ is the number of photos of property available during period t. We calculate $IMAGE\_QUALITY_{it}$ using machine learning techniques that are appropriate for the size of the data set (over 510,000 images).[5] We build a supervised image quality classifier that classifies each image as high quality (value of one) or low quality (value of zero). We calculate the average image quality, $IMAGE\_QUALITY_{it}$, of all the photos associated with property i in period t. For example, if property i had 10 images in period t and 8 images are labeled as high quality, then $IMAGE\_COUNT_{it} = 10$ and $IMAGE\_QUALITY_{it} = (8 \times 1 + 2 \times 0)/10 = 0.8$. Lastly, we include the distribution of photographed room types because professional photographers may have a more advanced understanding of which types of rooms will be most appealing to guests and thus, present more of these aspects of properties. Specifically, we compute the proportion of the photographs that depict each of five room types: bathroom, bedroom, kitchen, living room, and outdoor area. Then, for property i in period t, the distribution is represented by a vector: $\{BATHROOM\_PHOTO\_RATIO_{it}, BEDROOM\_PHOTO\_RATIO_{it}, KITCHEN\_PHOTO\_RATIO_{it}, LIVINGROOM\_PHOTO\_RATIO_{it}, OUTDOOR\_PHOTO\_RATIO_{it}\}$.

Table 1 presents a summary of the statistics for the key variables at the group level. To show the overall trends in the key variables, we report statistics for the pretreatment period (January 2016; i.e., when none of the properties in the sample were treated) and the posttreatment period (April 2017; when all the properties in the treatment group had been treated). As expected, image quality was stable over time in the control group (0.29 in January 2016 versus 0.30 in April 2017) and improved dramatically in the treatment group (0.27 in January 2016 versus 0.77 in April 2017), reflecting the high quality of photos shot by Airbnb professional photographers.

2.3. Analysis of Property Images

2.3.1. Classifying Images as High or Low Quality. We build a supervised deep learning algorithm that classifies images as high or low quality.

2.3.1.1. Training Set Construction. We chose a random sample of images from our data set to tag using AMT, a crowdsourcing platform for human tasks. A stratified (by the crude metric of quality) random sample is necessary to create a sample that is balanced as well as random. For each image, we asked the MTurkers (i.e., workers) to rate the image quality on a one to seven Likert scale, where one is "very bad" and seven is "excellent." For guidance, we provided examples of rated photos across the quality spectrum, and we explained what we meant by "quality" (e.g., the image should be "visually pleasing" and "clearly show room/house features"). Each image was evaluated by five qualified MTurkers. In the online appendix, we report details on image labeling using AMT. The image labels were further converted to a binary format ("high quality" versus "low quality"), following practices described in the computational aesthetics literature (Datta et al. 2006). The final training sample contained 1,155 high-quality images and 1,104 low-quality images.

2.3.1.2. Training Step.

2.3.1.2.1. CNN Approach. We apply a CNN, a deep learning framework widely applied in the field of computer vision with breakthrough performances on tasks including object recognition and image classification (Krizhevsky et al. 2012). Our CNN image quality classifier (see Figure 2) represents the architecture of a classic CNN model. The CNN consists of a sequence of neural layers (also called filters); the first layer extracts features from the input image and summarizes them into an intermediate output, which becomes the input with which the next layer generates a higher-level summary and so on. The key component in the CNN is a convolution kernel (or convolution filter), represented by an n by n weighting matrix. Given the intermediate output from the previous layer, the convolution kernel extracts features through a matrix dot product operation between the weighting matrix and the intermediate output. In other words, the sequence of layers in the CNN extracts a hierarchical set of image features from the input. A training set (in which the label "high quality" or "low quality" is already known for each image) enables the model to learn the relationships between the extracted features

Table 1. Summary of the Statistics of Airbnb Properties

	Control group		Treatment group	
	Mean	Standard deviation	Mean	Standard deviation
Sample size: # units (*N*)	7,211		212	
Pretreatment (January 2016)				
DEMAND (occupancy rate × 100)	31.07	35.93	32.57	32.30
# *RESERVATION DAYS* (demand component-days booked by guests)	5.49	8.79	6.62	8.4
# *AVAILABLE DAYS* (demand component-day not booked, not blocked)	12.91	12.34	14.87	11.86
# *BLOCKED DAYS* (demand component-days blocked by host)	12.60	13.42	9.51	12.43
IMAGE_QUALITY	0.29	0.27	0.27	0.25
IMAGE_COUNT	12.78	9.49	14.48	10.38
BATHROOM_PHOTO_RATIO	0.21	0.17	0.22	0.16
BEDROOM_PHOTO_RATIO	0.30	0.20	0.29	0.19
KITCHEN_PHOTO_RATIO	0.11	0.11	0.10	0.10
LIVINGROOM_PHOTO_RATIO	0.19	0.17	0.18	0.16
OUTDOOR_PHOTO_RATIO	0.19	0.20	0.20	0.20
SECURITY_DEPOSIT	181.97	347.75	202.77	333.07
CLEANING_FEE	48.02	53.04	54.69	58.95
MAX_GUESTS	3.19	2.11	3.50	2.23
SUPER_HOST	0.09	0.29	0.15	0.35
INSTANT_BOOK	0.11	0.31	0.11	0.31
MINIMUM_STAY	2.62	2.71	2.57	2.92
RESPONSE_RATE	91.00	16.29	92.25	14.31
RESPONSE_TIME (minutes)	261.07	365.66	225.12	338.39
NIGHTLY_RATE (average property price)	179.74	249.31	170.15	240.80
REVIEW_COUNT	16.41	26.25	20.56	26.98
HAS_RATING	0.68	0.47	0.78	0.41
RATING_OVERALL (overall score out of 100; summarizes multidimensional subratings)	92.88	7.14	93.99	5.04
RATING_COMMUNICATION	9.71	0.64	9.74	0.49
RATING_ACCURACY	9.50	0.76	9.54	0.56
RATING_CLEANLINESS	9.21	0.97	9.26	0.82
RATING_CHECKIN	9.67	0.67	9.72	0.48
RATING_LOCATION	9.39	0.82	9.40	0.70
RATING_VALUE	9.24	0.79	9.34	0.60
ZILLOW_RENTAL	2,459.95	800.40	2,474.75	845.05
UTILITY	0.18	0.03	0.17	0.04
INACTIVE PROPERTIES (proportion; a property in a month is *inactive* if it is blocked full month)	0.24		0.17	
Posttreatment (April 2017)				
DEMAND (occupancy rate × 100)	36.79	39.46	46.58	38.37
# *RESERVATION DAYS* (demand component-days booked by guests)	8.59	10.3	10.97	10.13
# *AVAILABLE DAYS* (demand component-day not booked, not blocked)	16.23	11.56	13.41	11.49
# *BLOCKED DAYS* (demand component-days blocked by hosts)	5.17	8.79	5.62	8.94
IMAGE_QUALITY	0.30	0.27	0.77	0.22
IMAGE_COUNT	16.22	11.98	19.61	11.53
BATHROOM_PHOTO_RATIO	0.21	0.16	0.21	0.15
BEDROOM_PHOTO_RATIO	0.30	0.19	0.30	0.18
KITCHEN_PHOTO_RATIO	0.11	0.10	0.09	0.10
LIVINGROOM_PHOTO_RATIO	0.19	0.15	0.18	0.17
OUTDOOR_PHOTO_RATIO	0.20	0.19	0.21	0.21
SECURITY_DEPOSIT	149.53	325.71	179.82	280.53
CLEANING_FEE	41.51	55.93	46.02	51.01
MAX_GUESTS	3.37	2.26	3.50	2.31
SUPER_HOST	0.19	0.39	0.29	0.45

Table 1. (Continued)

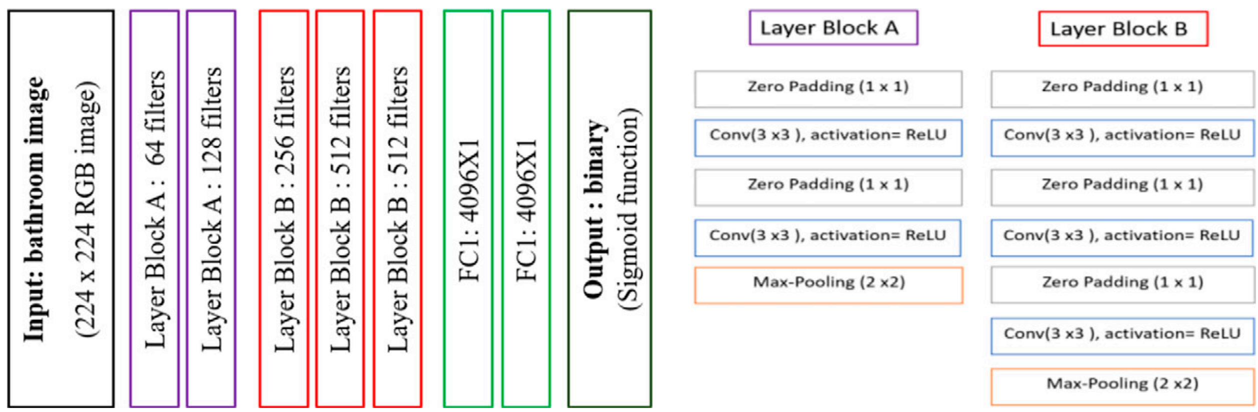
	Control group		Treatment group	
	Mean	Standard deviation	Mean	Standard deviation
INSTANT_BOOK	0.11	0.32	0.16	0.37
MINIMUM_STAY	2.36	3.75	2.56	4.06
RESPONSE_RATE	96.63	12.17	95.82	12.50
RESPONSE_TIME (minutes)	150.42	266.86	167.72	287.81
NIGHTLY_RATE (average property price)	237.97	319.49	210.14	146.81
REVIEW_COUNT (number of guest reviews)	32.89	49.33	41.80	46.37
HAS_RATING	0.82	0.38	0.84	0.36
RATING_OVERALL (overall score out of 100; summarizes multidimensional subratings)	93.80	5.34	94.16	4.45
RATING_COMMUNICATION	9.80	0.45	9.82	0.40
RATING_ACCURACY	9.60	0.60	9.66	0.51
RATING_CLEANLINESS	9.40	0.75	9.40	0.73
RATING_CHECKIN	9.79	0.45	9.78	0.45
RATING_LOCATION	9.47	0.67	9.50	0.65
RATING_VALUE	9.36	0.64	9.39	0.57
ZILLOW_RENTAL	2,459.80	822.06	2,426.22	823.28
UTILITY	0.18	0.04	0.17	0.04
INACTIVE PROPERTIES (proportion)	0.41		0.32	
Time invariant				
APARTMENT	0.67	0.47	0.60	0.49
ENTIREHOME	0.59	0.49	0.61	0.49
# BEDS	1.70	1.22	1.81	1.21
POOL	0.08	0.28	0.10	0.30
BEACH	0.01	0.10	0.02	0.14
AC	0.98	0.13	0.99	0.06
PARKING	0.44	0.50	0.50	0.50
INTERNET	0.98	0.13	0.99	0.09
TV	0.74	0.44	0.79	0.41
WASHER	0.59	0.49	0.60	0.49
MICROWAVE	0.09	0.29	0.15	0.36
ELEVATOR	0.23	0.42	0.20	0.40
GYM	0.10	0.29	0.11	0.31
FAMILY_FRIENDLY	0.23	0.42	0.19	0.39
SMOKE_DETECTOR	0.46	0.50	0.55	0.50
SHAMPOO	0.35	0.48	0.45	0.50

and the labels such that the model can extract the features that have the most discriminative power for predicting the labels. To reduce the overfitting problem in the training step, we increase the size of and random variation within the training sample by randomly applying one of three transformations to each image (e.g., data augmentation; see Krizhevsky et al. 2012): (1) flipping the input image horizontally, (2) rescaling the input image within a scale of 1.2, or (3) rotating the image within 20°.

2.3.1.2.2. Transfer Learning and Fine Tuning the Parameters of the CNN. Because a deep learning model has many filters and parameters, it requires a large quantity of data to train the model. We overcome our limited training data by leveraging transfer learning. We start with the pretrained parameters of the widely applied CNN model, VGG-16 (Visual Geometry Group-16 layer; trained on over 1 million images; Simonyan and Zisserman 2015), and fine tune the parameters with 80% of our training set of Airbnb property images. We use the remaining 20% as a holdout sample to test the performance of the trained CNN, and we achieve 90.4% accuracy. The high accuracy in predicting image quality ensures a valid interpretation of our results of image quality in the demand model. In the online appendix, we provide a detailed description of the CNN architecture and technical notes on the training process.

2.3.1.3. Prediction Step. After the CNN classifier learns the relationship between image features and the image label, it can be applied to the unlabeled images (Figure 2). The classifier takes the unlabeled image as the input, extracts the hierarchical set of image features using the parameters of the trained classifier, and outputs the predicted label: either "one" for a high-quality image or "zero" for a low-quality image.

Figure 2. (Color online) Description of Architecture and Layer Description of the CNN Classifier



Notes. Filters: The number of convolution windows (i.e., number of feature maps) on each convolution layer. Zero-padding: Pads the input with zeros on the edges to control the spatial size of the output. Zero-padding has no impact on the predicted output. Max-pooling: Subsampling method. A 2×2 window slides through (without overlap) each feature map at that layer, and the maximum value in the window is picked as the representation of the window. This reduces computation and provides translation invariance. ReLU, Rectified Linear Unit; FC, Fully Connected; RGB, Red-Green-Blue.

2.3.2. Classifying the Image Room Type.

We build another deep learning model to automatically classify the room in a property image as a bathroom, bedroom, kitchen, living room, or outdoor area. Again, we use transfer learning—we start with Places205, which was pretrained on a large scene classification data set (Zhou et al. 2014), and we fine tune the classifier on our own training data set (i.e., with known labels) that we collected from real estate-related websites on which a vast number of indoor/outdoor images are classified into categories: 54,557 images of "bathrooms," 59,082 images of "bedrooms," 88,030 images of "kitchens," 81,819 images of "living rooms," and 5,734 images of "outdoor areas." The trained classifier achieves 95.05% accuracy on the holdout sample. We then apply the trained classifier to the unlabeled property images. In the online appendix, we provide a description of the training set and technical notes for the training step of the room-type classifier.

3. Methods and Results

We implement a DiD analysis (Heckman et al. 1997) with PSW to mitigate the endogeneity concern.

3.1. DiD Analysis

The DiD analysis requires the identification of a treatment group and (comparable) control group. Our treatment group consists of the 221 properties that did not have verified photos in January 2016 but acquired at least one verified photo by April 2017. The control group consists of the 7,211 properties that had only unverified photos during the observation window.

In an ideal setting, the two groups would be comparable such that the impact of treatment (i.e., the adoption of verified photos) on demand would be reflected by the demand difference in the posttreatment period. In our unrandomized (i.e., self-selecting) setting, however, the treatment is endogenous, and the two groups may not be comparable. As shown in Table 1, the treatment and control groups differ on some pretreatment covariates. If certain differences affect both property demand and hosts' decisions about whether to join the photography program, then we cannot simply attribute any observed difference in the property demand to the treatment (Athey and Imbens 2006). To mitigate the endogeneity concern, we use the PSW method to create groups that are sufficiently comparable on the observed characteristics.

3.2. The PSW Method

The propensity score is the probability that an individual unit receives treatment, conditional on a set of observed covariates (Rosenbaum and Rubin 1983). Propensity scores can be used to "balance" samples and are widely used to make two groups comparable with respect to covariates (Rosenbaum 2002).

In practice, the true propensity scores are often unknown, so the PSW method estimates propensity scores by modeling the treatment probability as a function of the observed covariates. The propensity score of unit i, ps_i, is computed via a specified function, $ps_i = f(X_i\beta)$, where X_i is a $1 \times M$-dimensional vector of pretreatment observed covariates of unit i and β is an $M \times 1$-dimensional vector of parameters for X.

The model finds a set of parameters that maximizes the treatment likelihood in the sample data (Rosenbaum and Rubin 1983). We estimate the parameters via logistic regression because the treatment assignment is a binary variable (i.e., treatment or control). The selection of X is based on a covariate balance check (see the online appendix). The differences in the means of covariates between the treatment and control groups should be minimized. With parameter vector β estimated, we approximate for each unit i (with observed covariates X_i) the propensity score $\widehat{ps_i(X_i)}$, which is used to compute the sample weights.

In the X of the PSW model, we include covariates to control for the self-selection issue identified by Zhang et al. (2019)—that rational hosts should adopt high-quality images if they have an appropriately high-quality property and level of service. We control for the service quality through the *Response Rate* and *Response Time* (similar to Zhang et al. 2019) and capture the property quality with a rich set of property characteristics such as property amenities (internet, gym, beach, pool, etc.), property size and type, and host quality (super host).

3.2.1. Computing Sample Weights Based on Propensity Scores.

We use propensity scoring for a weighting strategy—the inverse probability of treatment weighting (IPTW) method—in the DiD analysis (Austin and Stuart 2015). The IPTW method calculates a weight w for unit i by taking the inverse of unit i's propensity score:

$$\omega_i(T, X_i) = \frac{T}{\widehat{ps_i(X_i)}} + \frac{1-T}{1-\widehat{ps_i(X_i)}}, \tag{1}$$

where $\widehat{ps_i(X_i)}$ is the estimated propensity score of unit i computed with its observed covariates X_i and T is a dummy variable that equals one if it is in the treatment group and zero otherwise.

The PSW results are validated to ensure that the treatment and control groups in the weighted sample are balanced on the included covariates (see the online appendix for details on the PSW approach, results, and validation). As shown in Table 2, the group means are statistically indifferent, suggesting that the PSW method successfully removed systematic differences in host and property observed characteristics that might confound the treatment assignment. See Section V.5 in the online appendix for the analysis details.

3.3. Model Specification and DiD Estimator

We obtain our DiD estimator through a weighted least squares regression, which computes sampling weights using estimated propensity scores. Let $DEMAND_{itcym}$ denote the demand for property i (in city c) in year y and month m (further, let t denote month m in year y):

$$\begin{aligned} DEMAND_{itcym} = {} & INTERCEPT + \alpha TREATIND_{it} \\ & + \lambda CONTROLS_{it} + PROPERTY_i \\ & + SEASONALITY_{cym} + \varepsilon_{it}, \end{aligned} \tag{2}$$

where $TREATIND_{it}$ is the treatment status indicator, which equals one if property i has received treatment in period t and equals zero otherwise. The key coefficient α estimates the percentage change in property demand that is associated with the treatment (i.e., having verified photos), and ε_{it} is a random shock in period t on property i's demand, assumed to follow an independent and identically distributed normal distribution. The vector $CONTROLS_{it}$ represents a set of control variables that may be correlated with property demand: for example, the property rules and consumer reviews.[6] We include the property fixed effect term, $PROPERTY_i$, to capture time-invariant factors that may impact property demand, such as geographic information and property-specific characteristics. We also include the time fixed effect term $SEASONALITY_{cym}$, which captures any city-specific trends in property demand (i.e., we allow each city to have its own seasonal pattern).

Note that the key outcome variable in this study is property demand, operationalized as the monthly

Table 2. PSW Validation: Covariate Balance Check

	Weighted means		*T* test	
Variables	Treated	Untreated	*t*	*p*-value
REVIEW_COUNT	20.56	19.88	0.27	0.790
IMAGE_QUALITY	0.27	0.25	1.00	0.316
IMAGE_COUNT	14.48	15.1	−0.67	0.506
NIGHTLY_RATE	170.15	191.36	−1.14	0.257
MINIMUM_STAY	2.57	2.57	−0.00	1.000
MAX_GUESTS	3.5	3.67	−0.82	0.410
RESPONSE_RATE	92.25	91.19	0.79	0.431
RESPONSE_TIME (minutes)	225.12	260.98	−1.18	0.238
SUPER_HOST	0.15	0.11	1.05	0.292
INSTANT_BOOK	0.11	0.11	−0.14	0.888
# BLOCKED DAYS	9.51	8.32	1.10	0.271
# RESERVATION DAYS	6.62	6.74	−0.16	0.877
PARKING	0.5	0.49	0.09	0.929
POOL	0.1	0.08	0.76	0.445
BEACH	0.02	0.02	0.34	0.737
INTERNET	0.99	1	−0.58	0.563
TV	0.79	0.81	−0.55	0.579
WASHER	0.6	0.57	0.81	0.419
MICROWAVE	0.15	0.13	0.64	0.523
ELEVATOR	0.2	0.21	−0.22	0.826
GYM	0.11	0.13	−0.69	0.490
FAMILY_FRIENDLY	0.19	0.2	−0.56	0.576
SMOKE_DETECTOR	0.55	0.52	0.62	0.534
SHAMPOO	0.45	0.44	0.09	0.929
BATHROOM_PHOTO_RATIO	0.22	0.21	1.05	0.295
BEDROOM_PHOTO_RATIO	0.29	0.29	−0.26	0.795
KITCHEN_PHOTO_RATIO	0.1	0.1	−0.15	0.880
LIVINGROOM_PHOTO_RATIO	0.18	0.19	−0.52	0.602
OUTDOOR_PHOTO_RATIO	0.2	0.21	−0.13	0.898
SECURITY_DEPOSIT	202.77	225.19	−0.65	0.517
# OPEN DAYS	21.49	22.68	−1.10	0.271
CANCELLATION_STRICT	0.26	0.29	−0.89	0.371
# BEDS	1.81	1.96	−1.17	0.242
APARTMENT	0.6	0.61	−0.27	0.786
ENTIRE_HOME	0.61	0.64	−0.64	0.522

occupancy rate. As an empirical extension, in the online appendix, we analyze whether the property price changed with treatment (i.e., we replace the dependent variable in Equation (2) with the property's nightly rate). We find that after controlling for seasonality, property characteristics, and time-varying variables such as the number of reviews, the coefficient of the treatment on property price is insignificant (see Section V.8 in the online appendix for a detailed discussion).[7]

As shown in Table 3, the estimated coefficient of the key variable *TREATIND* suggests a positive, significant treatment effect. Specifically, properties that used verified photos had 8.985% higher occupancy than the control group. On average, an untreated property was open 18.1 days per month, so the treatment coefficient corresponds to an average of 18.1 days/month × 8.985% × 12 months/year = 19.5 additional booked days in a year (or 1.6 additional days in a month). In terms of revenue, the acquisition of verified photos corresponds to an average of \$179.5/day × 19.5 days = \$3,500.30 more per year.[8]

3.4. Validating the DiD Model

We implement a set of analyses to validate our combination of the DiD model and the PSW strategy. We begin with a falsification check that examines the critical "common pretreatment trends" assumption, followed by a Rosenbaum bounds analysis for the selection on unobservables and several additional robustness checks.

3.4.1. Falsification Check: Pretreatment Trends.

The validity of the DiD approach (Equation (2)) relies on a critical assumption of common pretreatment trends; that is, the two weighted groups should exhibit common trends in their demand before treatment (Angrist and Pischke 2008). We test the common trends assumption using a relative time model of pretreatment periods. Following the extant literature (e.g., Wang and Goldfarb 2017), we decompose the pretreatment periods into a series of dummy variables:

$$\begin{aligned} DEMAND_{icym} = {} & INTERCEPT + \alpha TREATIND_{it} \\ & + \sum_j \beta_j (PRE_{it}(j) \cdot TREAT_i) \\ & + \lambda CONTROLS_{it} + SEASONALITY_{cym} \\ & + PROPERTY_i + \varepsilon_{it}. \end{aligned} \tag{3}$$

We define *PRE(1)* as the period prior to the treatment month and set it as the reference period (i.e., we normalize its coefficient to zero). Then, *PRE(2)* is two months prior to treatment, *PRE(3)* is three months prior to treatment, and *PRE(4)* represents all the periods

Table 3. DiD Model: Regressing Property Demand on Verified Photos

	Main DiD model (Equation (2))	
Variables	Estimates	Robust standard error
TREATIND	8.985***	1.660
log(*REVIEW_COUNT*$_{t-1}$)	9.375***	0.930
NIGHTLY_RATE	−0.146***	0.0320
INSTANT_BOOK	4.156**	1.361
CLEANING_FEE	0.0808***	0.0184
MAX_GUESTS	0.260	1.117
RESPONSE_RATE	0.0699	0.0430
RESPONSE_TIME (minutes)	−0.000477	0.00161
MINIMUM_STAY	0.133	0.131
SECURITY_DEPOSIT	0.00177	0.00201
SUPER_HOST	3.801*	1.494
BUSINESS_READY	1.806	0.985
CANCELLATION_STRICT	1.016	1.271
HAS_RATING	14.32	12.25
HAS_RATING × *COMMUNICATION*	−0.212	1.420
HAS_RATING × *ACCURACY*	0.878	1.211
HAS_RATING × *CLEANLINESS*	−1.344	1.133
HAS_RATING × *CHECKIN*	−2.060	1.526
HAS_RATING × *LOCATION*	−0.757	1.183
HAS_RATING × *VALUE*	2.141	1.176
Fixed effect	Property	
Seasonality	City-year-month	
Number of observations	76,901	
R^2	0.6608	

Notes. The demand model does not include the number of open nights because the dependent variable is computed as a function of it (occupancy rate = #reservation days/#open days). This variable is used in the matching step (see Table A1 in the online appendix).

*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

from the beginning (i.e., January 2016) through four months prior to treatment (Autor 2003). For properties with fewer than four pretreatment periods (e.g., a property that acquired verified photos in February 2016 would have only one pretreatment period), the period dummies that correspond to months before January 2016 are set to zero.

The set of coefficients β_j allows us to validate the DiD model by comparing the trends in the property demand of the weighted control and treatment groups prior to treatment. The common trends assumption is validated if there is no significant pretreatment difference between the groups, and Table 4 confirms that none of the β_j period dummy coefficients are statistically different from zero. Moreover, the set of β_j does not exhibit an increasing trend in the property demand for the treatment units compared with the control units. In other words, the demand for the treated units was not already deviating from the demand for the control units prior to treatment, which would suggest that the difference between groups was caused by an idiosyncratic shock that affected both the treatment likelihood and property demand rather than by the treatment itself.

3.4.2. Selection on Unobservables. The PSW method addresses endogeneity concerns regarding observed variables, but it does not rule out a possible hidden bias involving unobserved variables that might influence the treatment likelihood and outcome variable simultaneously. We assess the sensitivity of our estimation to a potential hidden bias with the Rosenbaum bounds test (Rosenbaum 2002), which determines how much of a change in the odds ratio of treatment would need to be because of unobservables to nullify the treatment effect identified by the PSW method. We can be more confident about the validity of the estimation results if the Rosenbaum bounds test suggests that unobservables would have to cause a large change in the odds ratio to overturn the estimated treatment effect.

Our results for the Rosenbaum bounds test with an examination of the Hodges–Lehmann estimates (Rosenbaum 1993) suggest that unobservables would need to increase the odds ratio of treatment by at least 55% (i.e., $\gamma \geq 1.55$) to overturn the positive estimated treatment effect on property demand. The results of our sensitivity analysis are similar to the results obtained in the extant literature (DiPrete and Gangl 2004, Sun

Table 4. Falsification Check: A Relative Time Model of Pretreatment Trends

	Relative time model (Equation (3))	
Variables	Estimates	Robust standard error
PRE(4) × *TREAT*	2.230	2.503
PRE(3) × *TREAT*	−0.894	3.164
PRE(2) × *TREAT*	1.302	3.102
PRE(1) × *TREAT* (reference month)	—	—
TREATIND	9.988***	2.311
Property (nonphoto) characteristics		
log($REVIEW\_COUNT_{t-1}$)	9.338***	0.933
NIGHTLY_RATE	−0.147***	0.0320
INSTANT_BOOK	4.166**	1.363
CLEANING_FEE	0.0811***	0.0184
MAX_GUESTS	0.199	1.108
RESPONSE_RATE	0.0730	0.0431
RESPONSE_TIME (minutes)	−0.000403	−0.00161
MINIMUM_STAY	0.132	0.131
SECURITY_DEPOSIT	0.00175	0.002
SUPER_HOST	3.764*	1.495
BUSINESS_READY	1.821	0.985
CANCELLATION_STRICT	1.05	1.271
HAS_RATING	14.57	12.22
HAS_RATING × *COMMUNICATION*	−0.229	1.421
HAS_RATING × *ACCURACY*	0.978	1.21
HAS_RATING × *CLEANLINESS*	−1.403	1.134
HAS_RATING × *CHECKIN*	−2.117	1.521
HAS_RATING × *LOCATION*	−0.716	1.182
HAS_RATING × *VALUE*	2.118	1.177
Fixed effect	Property	
Seasonality	City-year-month	
Number of observations	76,901	
R^2	0.6609	

*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

and Zhu 2013, Manchanda et al. 2015, Li et al. 2016), which reported γ values ranging from 1.2 to 1.6. This suggests that our study is robust to the hypothetical unobserved factors that may affect the treatment likelihood. We provide detailed methods and results in the online appendix.

3.4.3. Additional Robustness Tests. In addition to the falsification test (Section 3.4.1) and the Rosenbaum analysis (Section 3.4.2), we perform a comprehensive set of seven tests that verify the robustness of our main findings. We briefly describe the tests; detailed descriptions and results are in Section V in the online appendix.

First, we run two analyses to test for potential inflation in the treatment coefficient in the long term, which could occur if Airbnb's ranking algorithm favors properties with verified images. We estimate our main DiD specification on a set of subsamples in which we include a shorter period (four months post-treatment) for each treated unit. The estimated coefficient of *TREATIND* remains significant and positive. We also use a relative time model to estimate the treatment effect in each month following treatment. We find that the estimated coefficient of *TREATIND* in the month following treatment is close to our main finding. The two analyses add confidence that our main finding was not driven by long-term inflation.

Second, we test whether properties with particular amenities were more likely to acquire verified photos at a time when these amenities were most attractive (e.g., a pool in the summer). We include interaction terms for the dummy *AFTER* and meaningful amenities (pool, beach, air conditioning) in the DiD model and find consistent estimated results. The results suggest that the positive treatment coefficient is not because of higher demand for these amenities following treatment or in particular seasons.

Third, we investigate whether the positive treatment effect was driven by unobserved enhancements of the property or host. We regress the multidimensional review ratings (communication, cleanliness, etc.) on *TREATIND*. The coefficient of *TREATIND* is statistically insignificant, suggesting no substantial change in the quality of guests' experiences after the acquisition of verified photos.

Fourth, we examine whether hosts in the treatment group changed their management of the property calendar (e.g., made the property open for more nights per month) at the same time as they adopted verified photos. We regress *#_OPEN_DAYS* on *TREATIND* and other controls and find an insignificant coefficient of *TREATIND*, suggesting that the main results are not because of an increase in property availability after treatment. Additionally, we include *#_OPEN_DAYS* in the demand model and have similar finding of estimated *TREATIND*.

Fifth, we replicate the DiD model with a log transformation of the numeric variables and find no change in the estimated treatment coefficient.

Sixth, Zhang et al. (2019) find that image quality affects the review probability, and the number of reviews affects property demand. We incorporate the probability that a guest writes a review for the property. In this test, we match properties based on review probability as well as other covariates. The DiD estimation results on the matched sample are consistent with our main results and the estimated coefficient of review probability.[9]

Seventh, we use a different control group: the pretreated units (i.e., those that adopted verified photos before January 2016 and thus, were excluded from the main analyses). The pretreated units may represent a stronger control because the pretreated and to-be-treated properties might be more similar in the characteristics that drive the treatment likelihood (Narang and Shankar 2019). Two analyses report estimated coefficients of *TREATIND* that are consistent with our main finding.

3.5. Examining the Aesthetic Qualities of Verified vs. Unverified Photos

Our main analyses in Section 3.3 established that the demand for properties with verified images was 8.98% higher than the demand for properties with unverified images. In this section, we explore the potential sources of the positive coefficient of verified photos *correlationally* by extracting image aesthetic quality. We analyze the images and then estimate a demand equation (see Equation (4)) that captures three aspects of a property's image set: *IMAGE_COUNT* (the number of property images), *IMAGE_QUALITY* (the average of the binary labels of "high quality" (one) and "low quality" (zero)), and the distribution of the five main types of photographed rooms. In Section 2.3 and the online appendix, we define the variables and provide technical details regarding their measurements:

$$\begin{aligned} DEMAND_{itcym} = {} & INTERCEPT + \alpha TREATIND_{it} \\ & + \mu IMAGE\_COUNT_{it} \\ & + \gamma IMAGE\_QUALITY_{it} \\ & + \rho_1 BATHROOM\_PHOTO\_RATIO_{it} \\ & + \rho_2 BEDROOM\_PHOTO\_RATIO_{it} \\ & + \rho_3 KITCHEN\_PHOTO\_RATIO_{it} \\ & + \rho_4 LIVING\_PHOTO\_RATIO_{it} \\ & + \lambda CONTROLS_{it} + SEASONALITY_{cym} \\ & + PROPERTY_i + \varepsilon_{it}. \end{aligned} \tag{4}$$

Table 5 reports the estimation results; the model specification does not include *IMAGE_QUALITY* in column (1) and does include it in column (2). Recall that

Table 5. DiD Model: Controlling for the Number and Quality of Property Images

	(1) Without image quality		(2) Including image quality	
Variables	Estimates	Robust standard error	Estimates	Robust standard error
TREATIND	7.453***	1.777	4.397*	2.190
Property (nonphoto) characteristics				
log(*REVIEW_COUNT*$_{t-1}$)	9.754***	0.944	9.570***	0.942
NIGHTLY_RATE	−0.183***	0.0325	−0.187***	0.0325
INSTANT_BOOK	3.768**	1.357	3.664**	1.349
CLEANING_FEE	0.0931***	0.0188	0.0955***	0.0187
MAX_GUESTS	0.247	1.098	0.285	1.094
RESPONSE_RATE	0.0868*	0.0427	0.0886*	0.0427
RESPONSE_TIME (minutes)	−0.000232	0.00159	−0.000151	0.00160
MINIMUM_STAY	0.172	0.132	0.171	0.133
SECURITY_DEPOSIT	0.00211	0.00201	0.00210	0.00200
SUPER_HOST	3.999**	1.495	3.890**	1.494
BUSINESS_READY	1.805	0.977	1.813	0.974
CANCELLATION_STRICT	1.282	1.277	1.308	1.278
HAS_RATING	15.82	12.27	14.96	12.28
HAS_RATING × *COMMUNICATION*	0.0499	1.418	0.0702	1.423
HAS_RATING × *ACCURACY*	0.588	1.209	0.681	1.206
HAS_RATING × *CLEANLINESS*	−1.214	1.133	−1.144	1.141
HAS_RATING × *CHECKIN*	−2.260	1.525	−2.267	1.516
HAS_RATING × *LOCATION*	−0.785	1.186	−0.745	1.193
HAS_RATING × *VALUE*	2.124	1.176	2.023	1.182
AFTER × *POOL*	5.841	4.610	6.053	4.608
AFTER × *BEACH*	−11.29	10.91	−10.39	10.66
AFTER × *AC*	0.331	3.978	−0.198	3.963
Property image characteristics $^{(+)}$				
log(*IMAGE_COUNT*)	6.874***	1.724	5.518**	1.777
BATHROOM_PHOTO_RATIO	0.777	8.457	0.779	8.439
BEDROOM_PHOTO_RATIO	−2.575	7.539	−2.128	7.499
KITCHEN_PHOTO_RATIO	18.04	11.27	17.20	11.22
LIVINGROOM_PHOTO_RATIO	−12.07	8.430	−12.41	8.405
IMAGE_QUALITY			8.984**	3.371
Fixed effect	Property		Property	
Seasonality	City-year-month		City-year-month	
Number of observations	76,901		76,901	
R^2	0.6623		0.6628	

Note. The coefficient for *OUTDOOR_PHOTO_RATIO* is not estimated.
$^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$.

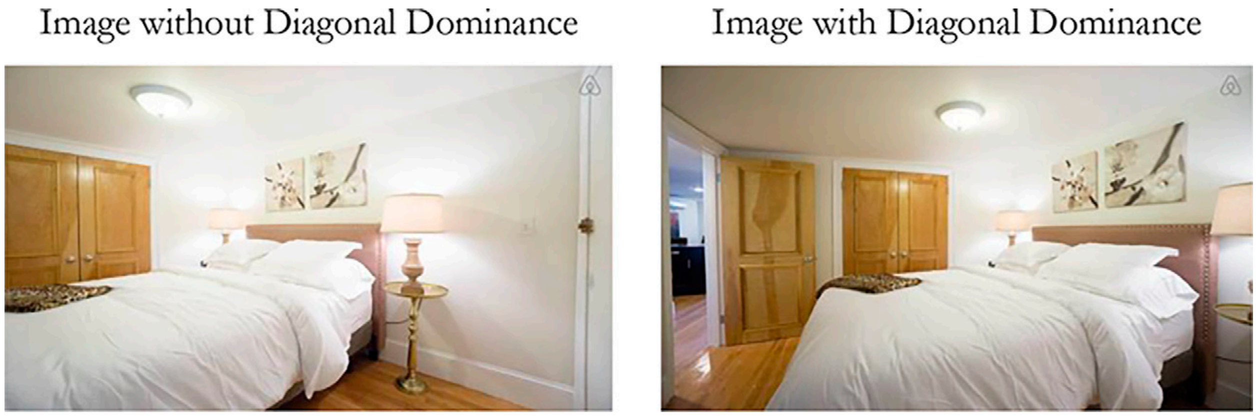
the estimated coefficient of the key variable *TREATIND* was 8.985 in the original model; the coefficient decreases to 7.453 in column (1) with the inclusion of *IMAGE_COUNT*, which has a positive and significant coefficient (none of the room-type ratios have a significant coefficient). The treatment coefficient decreases by another 41% (from 7.453 to 4.397) in column (2) with the inclusion of *IMAGE_QUALITY*, which has a positive and significant coefficient. This significant reduction suggests that the high quality of the verified images explains some but not all of the treatment effect, as there is a significant residual treatment coefficient even when we control for both covariates.

3.6. What Makes a Property Image Appealing?

Our CNN classifier predicted image quality with high accuracy, but the CNN-generated features are uninterpretable. That is, the features do not provide any insights for managers or photographers regarding how to improve their images based on CNN-generated features. Furthermore, Airbnb property images may be appealing for more reasons than image quality. Image quality captures only the vertical differentiation among images, but images can also differ horizontally in terms of taste. In an effort to provide actionable insights, we evaluate the relationship between property demand and 12 key image attributes identified in the photography literature. We first explain these features and then theorize their impact on property demand.

3.6.1. Interpretable Image Features. Multiple image attributes may affect consumers' perceptions and choices. To investigate which image attributes correlate

Figure 3. (Color online) Comparing Images on Diagonal Dominance



most strongly with Airbnb property demand, this section defines 12 key dimensions along which photographs can be compared and categorized. We use the art and photography literature to define the image attribute dimensions that are most relevant to property photos. Then, drawing from studies in this literature that pertain to the role of images in viewer perception, we theorize how each attribute could affect property demand.

The photography literature highlights 12 image attributes across three components: composition, color, and the figure-ground relationship (Datta et al. 2006, Freeman 2007, Wang et al. 2013). The features affect both the vertical quality of the photographs and horizontal qualities that might affect a property's appeal to potential guests who browse the images.

3.6.1.1. Component: Composition. Composition is the way in which visual elements are arranged, which guides the viewer's eyes to certain elements of the photograph (Freeman 2007). Expert photographers compose images that enable viewers to quickly identify the element that is the center of focus (Grill and Scanlon 1990). The most appropriate compositional attribute depends on the context; the following three compositional techniques are relevant for real estate photography.

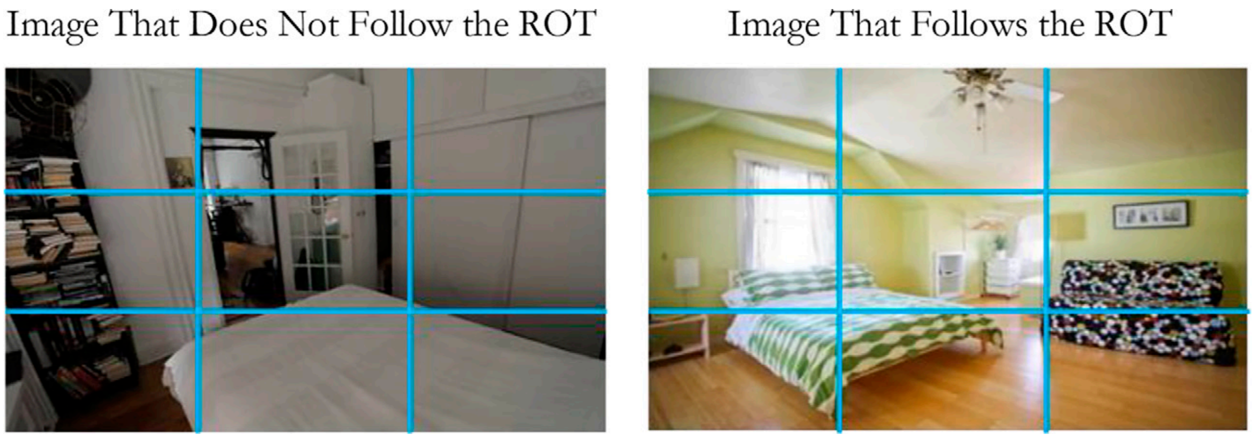
3.6.1.1.1. Attribute 1: Diagonal Dominance. A photographer can guide the viewer's eyes with leading lines, and in a rectangular frame, the two diagonals of the photograph are the longest possible straight lines. In a diagonally dominant photograph, the most salient visual elements are placed close to the two diagonals (Grill and Scanlon 1990). Diagonal dominance creates a perception of spaciousness, so we predict a positive relationship between diagonal dominance and property demand. In Figure 3, it is likely that viewers will perceive that the image in the right panel represents a larger room than the one in the left panel.

3.6.1.1.2. Attribute 2: Rule of Thirds. An image can be divided into nine equal parts with (imaginary) horizontal and vertical third lines. The ROT states that the main visual elements should be placed along the imaginary third lines or close to the four intersections (Krages 2005). These off-center focal points introduce movement into the photograph, engaging the viewer and making the image aesthetically pleasing and dynamic (Meech 2004), so we predict a positive relationship between the use of the ROT and property demand. In Figure 4, the image in the right panel follows the ROT better than the image in the left panel. Hence, when looking at the image, the viewer's attention first goes to the bed and then, its counterpoint—the other vertical third line. By contrast, in the image in the left panel, the focus and key objects are not obvious.

3.6.1.1.3. Attributes 3 and 4: Visual Balance of Intensity and Visual Balance of Color. Visual balance is the symmetry of visual elements (in this case, intensity and color) within an image (Krages 2005). The more symmetrical the elements within the image, the greater the visual balance, with the extreme case being perfect symmetry. Humans subconsciously consider visual balance to be aesthetically pleasing, and symmetry increases visual interest (Arnheim 1974, Bornstein et al. 1981). Visually balanced real estate images give viewers a feeling of order and tidiness, minimizing the cognitive demand required to process the images (Kreitler and Kreitler 1972, Machajdik and Hanbury 2010). We predict a positive relationship between visual balance and property demand. In Figure 5, the image in the right panel is more visually balanced than the one in the left panel, so the image in the right panel can be processed very quickly and provides a sense of order and cleanliness.

3.6.1.2. Component: Color. Color, one of the most significant elements in photography, affects the viewer's

Figure 4. (Color online) Comparing Images on the ROT



emotional arousal. Building on past research, Gorn et al. (1997) identified two dimensions of arousal: from boredom to excitement and from tension to relaxation. Excitement is preferred to boredom, and relaxation is preferred to tension. Three dimensions of color—hue, saturation (chroma), and brightness (value)—can affect the level of arousal. Each dimension has been widely studied in the marketing literature, particularly in the contexts of web design, product packaging design, and advertisement design (Gorn et al. 2004, Miller and Kahn 2005). We add another attribute, image clarity, that is affected by the combination of the aforementioned three.

3.6.1.2.1. Attribute 5: Warm Hue. Hues (e.g., red, green) are believed to be a major driver of emotion. The warmth in an image is affected by the actual colors of the subject, but the photographer can also manipulate hue by warming up or cooling down a picture during postprocessing. Warm hues (such as red and yellow) elicit higher levels of excitement (Valdez and Mehrabian 1994, Gorn et al. 2004), whereas cool hues (such as blue and green) elicit higher levels of relaxation. Hence, we predict a positive relationship between warm hues and property demand. In Figure 6, we present a cool photo of a living room in the left panel and a warm photo of a living room in the right panel.

3.6.1.2.2. Attribute 6: Saturation. Saturation refers to the richness of color. Highly saturated images are colorful, whereas weakly saturated images contain low levels of pigmentation. Saturation is positively associated with happiness and purity; less saturated colors are associated with sadness and distress (Valdez and Mehrabian 1994, Gorn et al. 2004). Thus, we predict that real estate images with saturated colors can induce positive emotions in viewers and that there is a positive relationship between saturation and property demand. To illustrate the difference in emotional arousal, Figure 7 displays two images of the same room with different levels of saturation.

Figure 5. (Color online) Comparing Images on Visual Balance

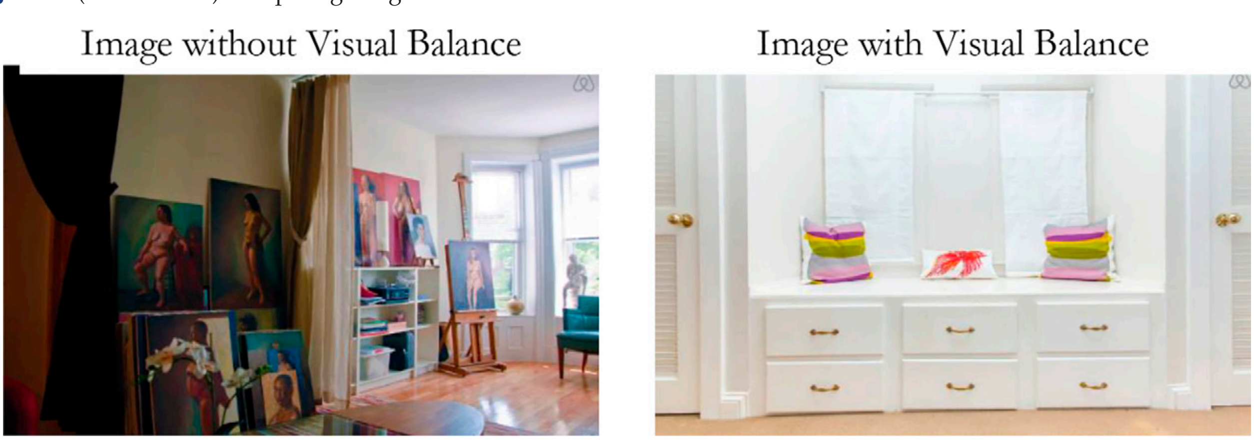
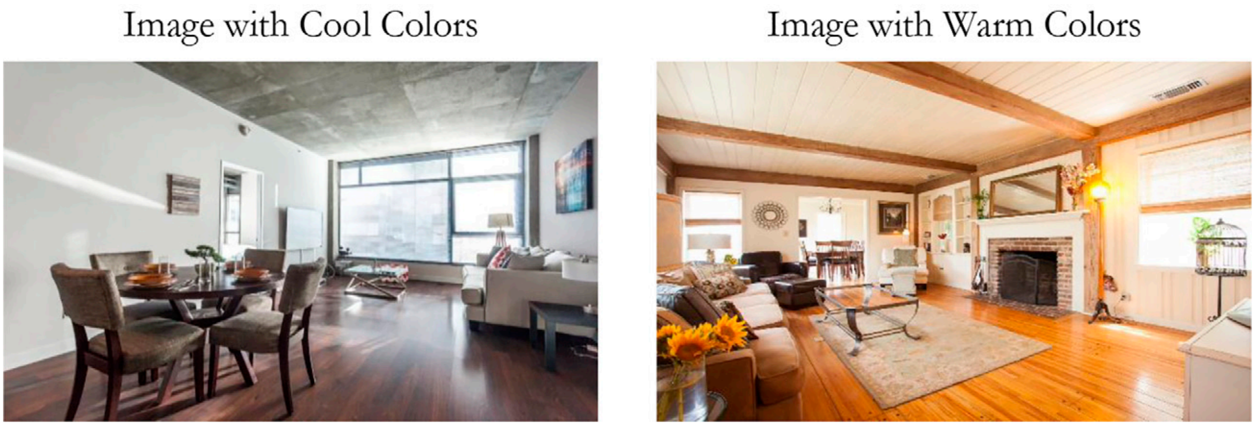


Figure 6. (Color online) Comparing Images on Warm Hue



3.6.1.2.3. Attributes 7 and 8: Brightness and the Contrast of Brightness. The photography literature identifies two image attributes regarding image illumination: brightness and its contrast. Brightness is the level of overall illumination; viewers prefer bright images as they induce a sense of relaxation but do not affect the level of excitement (Valdez and Mehrabian 1994, Gorn et al. 1997). Furthermore, sufficient illumination makes the content of an image clear to viewers because images convey information through pixel brightness; therefore, we predict a positive relationship between brightness and property demand. Meanwhile, the contrast of brightness is the variance in the illumination and describes whether the illumination is evenly distributed over the image with a smooth flow. In other words, a low contrast of brightness indicates an even distribution of illumination across the photograph. An uneven distribution (i.e., high contrast) of brightness may induce a feeling of harshness, so we predict a negative relationship between the contrast of brightness and property demand. For example, in Figure 8, the image in the right panel has a higher level of brightness and more uniform illumination (i.e., lower contrast of brightness) than the image in the left panel.

3.6.1.2.4. Attribute 9: Image Clarity. Clarity reflects the intensity of hues in the hue, saturation, and value space (Levkowitz and Herman 1993). An image is "dull" if it is dominated by desaturated colors or has near-zero hue intensities in some color channels (He et al. 2011). Amateur photographers often shoot dull photos, inducing a so-called haze effect in which some parts of the image look unclear and ill focused. By contrast, images with "clear" color reduce the friction in information transfer, so we predict a positive relationship between image clarity and property demand. In Figure 9, the photo in the right panel has higher clarity than the one in the left panel.

Figure 7. (Color online) Comparing Images on Saturation

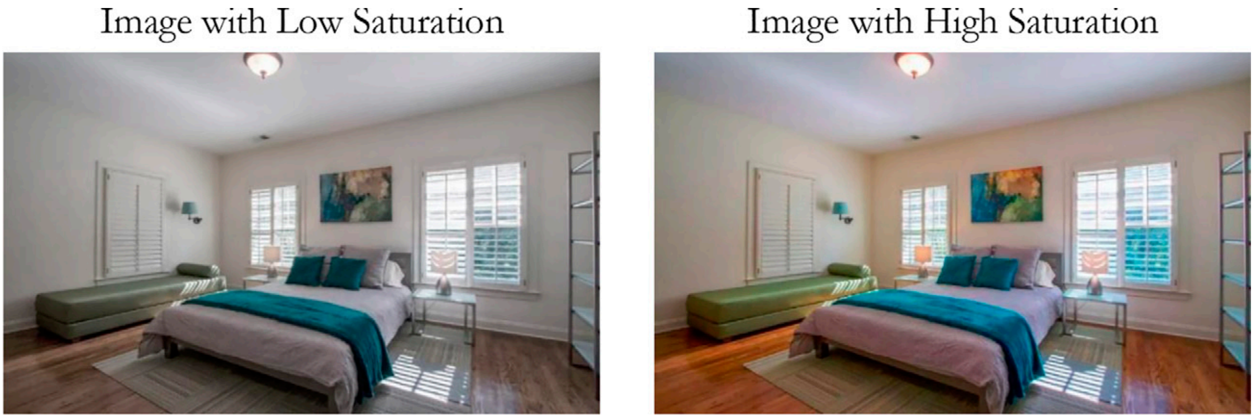
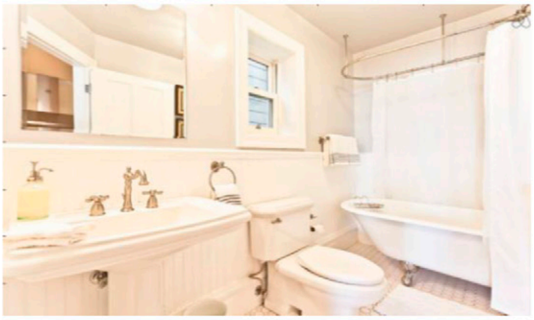


Figure 8. (Color online) Comparing Images on Illumination (Brightness and Its Contrast)



3.6.1.3. Component: The Figure-Ground Relationship.

3.6.1.3.1. Attributes 10, 11, and 12: Area Difference, Color Difference, and Texture Difference. The FG relationship within an image is evaluated in relation to three aspects—area, color, and texture. The principle of the figure-ground relationship is one of the most basic laws of perception and is used extensively by expert photographers to plan their photographs. In visual art, the *figure* refers to the key region (i.e., foreground), and the *ground* refers to the background. The FG relationship describes the separation between the figure and ground. Gestalt theory states that objects that share visual characteristics, such as size, color, and texture, are seen as belonging together (Arnheim 1974). Hence, the figure is more salient if it differs from the ground in size, color, and texture. In advertising research, consumers pay more attention to images with clear FG relationships (Larsen et al. 2004, Schloss and Palmer 2011), and we predict a positive relationship between FG separation and property demand. Figure 10 presents one set of images in which the figure is clearly separable from the ground and another set in which the separation is not obvious.

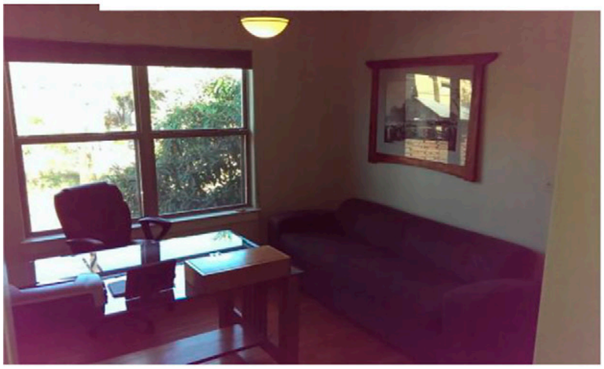
3.6.2. Measurement of Image Attributes and Related Statistics. Table 6 summarizes and briefly describes the 12 attributes.

3.6.2.1. Statistics of Image Attributes. We measure the 12 attributes using computer vision algorithms; detailed methods appear in the online appendix. In general, an image processing task involves the segmentation of an image into patches, detection of salient regions, and computation involving those regions. We compare the attributes across three groups of property images.

Group LQ (Low Quality). All low-quality images (all unverified; $N = 368{,}626$)

Group HQ_UN (Low Quality Unverified). All high-quality unverified images ($N = 69{,}380$)

Figure 9. (Color online) Comparing Images on Clarity



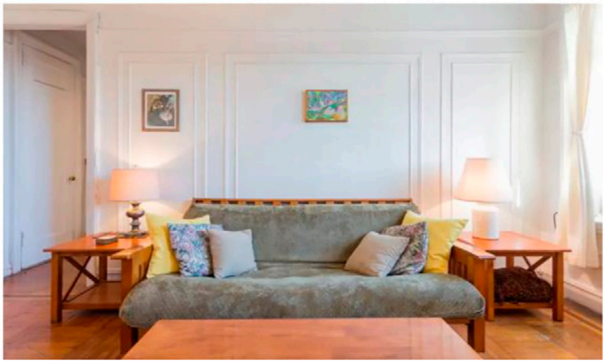
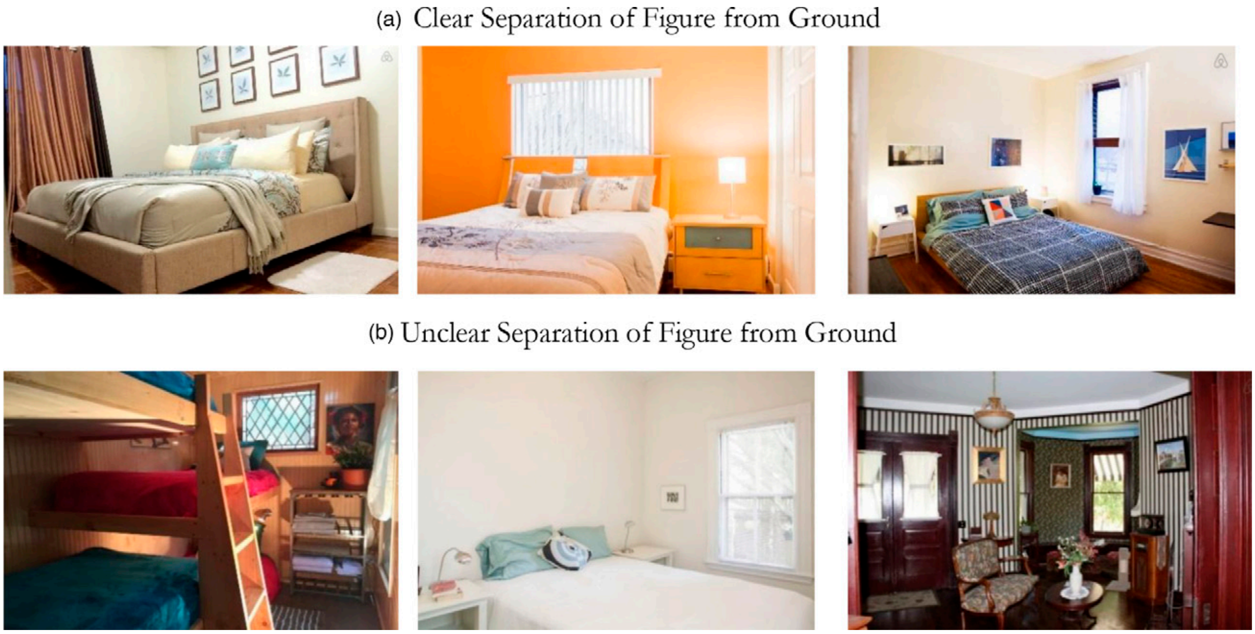


Figure 10. (Color online) Comparing Images on the Figure-Ground Relationship



Notes. (a) Clear separation of figure from ground. (b) Unclear separation of figure from ground.

Group HQ_V (High Quality Verified). All verified images (all high quality; $N = 72{,}608$)

The columns LQ, HQ_UN, and HQ_V of Table 7 present the means (standard deviations) for the image attributes by group. The HQ_V versus HQ_UN (rightmost) column presents the difference between the means of the HQ_UN and HQ_V groups along with the two-sample t statistics in parentheses; statistically significant differences (at a 5% significance level) are bolded.

The low-quality images rate lower than the high-quality images on all image attributes except for the contrast of brightness, which is theorized to have a negative relationship with the property demand. More interestingly, the unverified high-quality images also perform significantly worse than the verified high-quality images on most attributes. We conclude that there is a systematic difference in the high-quality images taken by Airbnb photographers versus those by other photographers.

3.6.3. Relating the Interpretable Image Attributes to Property Demand.

The 12 human-interpretable attributes have well-studied relationships with image quality, but the tastes of Airbnb consumers do not necessarily align perfectly with image quality. The lack of exogenous variation in the 12 attributes prevents us from drawing causal conclusions. Nevertheless, the DiD analysis may offer valuable exploratory insights. Table 8 presents the results[10] from the estimation of Equation (5):

$$\begin{aligned} DEMAND_{itcym} = {} & INTERCEPT + \alpha TREATIND_{it} \\ & + \mu IMAGE\_COUNT_{it} \\ & + \rho_1 BATHROOM\_PHOTO\_RATIO_{it} \\ & + \rho_2 BEDROOM\_PHOTO\_RATIO_{it} \\ & + \rho_3 KITCHEN\_PHOTO\_RATIO_{it} \\ & + \rho_4 LIVING\_PHOTO\_RATIO_{it} \\ & + \eta IMAGE\_ATTRIBUTES_{it} \\ & + \lambda CONTROLS_{it} + SEASONALITY_{cym} \\ & + PROPERTY_i + \varepsilon_{it}. \end{aligned} \tag{5}$$

Recall that we theorized that the contrast of brightness has a negative effect on property demand, and the remaining 11 image attributes have positive effects. The coefficients in Table 8 are of the expected sign: that is, significantly negative for the contrast of brightness and significantly positive for the other attributes. We can conclude that, controlling for all other variables, the 12 image attributes are significantly correlated with property demand.

Among the four composition attributes, the largest coefficients belong to the visual balance of color and visual balance of intensity. Both features describe the symmetry of the image, which is determined by both the physical arrangement of elements in the photograph and the photographer's position from which the image is taken.

Table 6. The 12 Image Attributes and Their Descriptions

Component		Attribute	Description
Composition	1	Diagonal dominance	Alignment of the key objects with the diagonals
	2	Visual balance-intensity	Symmetry of key objects around the imaginary vertical central line
	3	Visual balance-color	Symmetry of colors around the imaginary vertical central line
	4	Rule of thirds	Alignment of the key objects with the intersections of two imaginary horizontal and two vertical lines
Color	5	Warm hue	Dominance of warm colors (e.g., red and yellow) over cold colors (e.g., blue and purple)
	6	Saturation	The richness/vividness of image colors
	7	Brightness	Overall level of illumination
	8	Contrast of brightness	The variance in the illumination flow across the image
	9	Image clarity	Whether image colors have sufficient intensity
Figure-ground relationship	10	Size difference	Difference in area between the figure and ground
	11	Color difference	Difference in color between the figure and ground
	12	Texture difference	Difference in texture between the figure and ground

Among the five color attributes, the largest magnitudes of coefficients belong to image clarity and the contrast of brightness. Image clarity is obviously desirable; images that do not convey information clearly cannot make elements of the property appear attractive to the consumer. One would expect most skilled photographers to prioritize image clarity, but Table 7 shows a significant difference in image clarity between the unverified and verified high-quality images; unsurprisingly, the verified photos score almost twice as high as the unverified low-quality images. Meanwhile, a low value is usually preferred for the contrast of brightness, although several hosts have complained (on the Airbnb host forums) that the contrast of brightness is so low in the verified photos that they appear washed out, which might drive guests away.[11] Our results do not find evidence to support this concern. The lower contrast of brightness in the verified photos (Table 7) may actually make property images more appealing to viewers.

Among the three FG relationship features, the most prominent one is the size difference, as suggested by its largest coefficient. When the figure has a much larger size than the background, it is better able to capture the viewer's attention. A photographer's ability to manipulate the size difference may be physically constrained; for instance, if there is little size difference between the two, it will be hard to separate the figure from the ground.

With the inclusion of the 12 image attributes, the coefficient of the key variable *TREATIND* falls to 1.721 and is statistically insignificant (Table 7), suggesting that the treatment effect is associated with the ability of Airbnb professional photographers to achieve more optimal values of the 12 interpretable attributes.

4. Discussion and Conclusions

This study employs a DiD analysis combined with the PSW method to investigate Airbnb's photography program. We find that the properties that acquired verified photos had an 8.98% higher occupancy rate, translating into an average increase of \$3,500.30 in annual revenue, than the properties without verified photos. The treatment coefficient decreased by 41% when we controlled for image quality, suggesting that a sizeable portion (but not all) of the coefficient of verified photos on property demand is attributable to the high quality of the professional images.

Furthermore, we shed light on what makes a good image for an Airbnb property. We show that the

Table 7. Summary Statistics: Mean (Standard Deviation) of Image Attributes of Verified vs. Unverified High-Quality Images

		LQ (low quality, 368,626 observations)	HQ_UN (high quality unverified, 69,380 observations)	HQ_V (high quality verified, 72,608 observations)	HQ_V vs. HQ_UN
Component	Image attribute	Mean(standard deviation)	Mean (standard deviation)	Mean (standard deviation)	Difference (t statistic)
Composition	Diagonal dominance	−0.342 (0.160)	−0.281 (0.109)	−0.236 (0.081)	**0.045***** (88.56)
	Visual balance of intensity	−0.865 (0. 110)	−0.774 (0.103)	−0.757 (0.105)	**0.017***** (30.78)
	Visual balance of color	−59.281 (19.460)	−53.093 (15.509)	−50.096 (15.070)	**2.997***** (36.93)
	Rule of thirds	−0.147 (0.082)	−0.089 (0.045)	−0.089 (0.047)	0.0003 (1.23)
Color	Warm hue	0.738 (0.230)	0.751 (0.208)	0.789 (0.181)	**0.038***** (36.77)
	Saturation	59.023 (37.528)	73.942 (31.300)	73.683 (26.929)	−0.259 (0.87)
	Brightness	136.029 (32.488)	154.212 (27.558)	175.802 (22.593)	**21.590***** 161.75)
	Contrast of brightness	60.601 (13.628)	58.029 (13.056)	53.996 (12.990)	**−4.033***** (58.33)
	Image clarity	0.324 (0.232)	0.413 (0.217)	0.595 (0.195)	**0.182***** (166.38)
Figure-ground relationship	Size difference	−0.405 (0.181)	−0.181 (0.188)	−0.140 (0.153)	**0.041***** (45.16)
	Color difference	23.090 (20.056)	33.054 (17.552)	39.063 (15.580)	**6.009***** (68.29)
	Texture difference	0.043 (0.033)	0.057 (0.026)	0.059 (0.018)	**0.002***** (16.92)

Notes. Standard deviations are in parentheses for the first three columns; t statistics are in parentheses for the rightmost column. Note that the composition measurements are negative because they reflect distances, and we subtract all distances from zero to preserve the absolute magnitude when the direction is reversed. A higher value (i.e., less negative) suggests better performance on that composition attribute. For example, a higher value of diagonal dominance suggests that the image is more diagonally dominant. Statistically significant differences (at a 5% significance level) are in bold.

***$p < 0.001$.

verified and unverified photos differ significantly on 12 interpretable image attributes identified by the photography literature. We theorize the impact of each of the 12 attributes on property demand, and we find supportive evidence for all 12 hypothesized relationships. We cannot draw causal conclusions from our observational study, but our findings nevertheless may inform content engineering strategies among Airbnb photographers, hosts who take their own photographs, and real estate marketers more broadly.

We use the PSW method to mitigate the effect of self-selection on our DiD analysis. There are, however, possible reasons that may cause risk to the PSW and DiD analyses; in a related study, motivated by our findings and leveraging the foundations of our model, Zhang et al. (2019) investigate the reason as to why a large proportion of Airbnb hosts does not adopt high-quality images even when Airbnb offers them for free. They find that high-quality images lead to higher expectations for the property and level of service. If guests hold inappropriately high expectations, they are more likely to be dissatisfied and less likely to write a review, if such expectations are not met. In the long run, a property may face lower demand from its relatively low number of accumulated reviews. That is, the service quality and property quality affect hosts' verified photo adoptions and hence, beside other factors, create a self-selection issue for our study. As discussed in Section 3.2, our PSW model helps to mitigate the issue through controlling for *Response Rate* and *Response Time* and a rich set of property characteristics. Additionally, note that Zhang et al. (2019) finds that the direct effect of image quality on demand is always positive, in the short and long term for all properties (consistent with our long-term robustness test; see Section V.7 of the online appendix). The negative effect of images' quality on property demand comes *indirectly* through a lower rate of review accumulation for properties using high-quality images. Similar to Zhang et al. (2019), we control for the number of reviews in the demand model.

The PSW approach accounts for and matches properties on observed characteristics, but unobserved aspects of property quality may also affect both the property demand and treatment likelihood. We assess the robustness of our results to unobserved factors with two follow-up analyses, a sensitivity analysis (Rosenbaum bounds test; Section 3.4.2) and a DiD estimation with the pretreated units (i.e., those that acquired verified photos before January 2016) as the control group, such that any unobservables that drive self-selection should affect the treatment and control groups equally (Section 3.4.3 and Section V of the online appendix). The results increase our confidence that the PSW method adequately mitigated the endogeneity concern.

There are a few limitations to this research. First, our trained CNN classifier has a relatively high

Table 8. DiD Model: Controlling for Interpretable Image Attributes

Component		Variables	Estimates	Robust standard error
		TREATIND	1.721	6.575
		Property (nonphoto) characteristics		
		log(*REVIEW_COUNT*)	9.279***	0.927
		NIGHTLY_RATE	−0.194***	0.0325
		INSTANT_BOOK	3.245*	1.351
		CLEANING_FEE	0.0955***	0.0185
		MAX_GUESTS	0.159	1.093
		RESPONSE_RATE	0.0946*	0.0424
		RESPONSE_TIME (minutes)	−0.000203	0.00158
		MINIMUM_STAY	0.159	0.130
		SECURITY_DEPOSIT	0.00189	0.00208
		SUPER_HOST	3.614*	1.518
		BUSINESS_READY	2.182*	0.971
		CANCELLATION_STRICT	1.670	1.256
		HAS_RATING	14.02	11.76
		HAS_RATING × *COMMUNICATION*	0.0507	1.393
		HAS_RATING × *ACCURACY*	0.156	1.200
		HAS_RATING × *CLEANLINESS*	−0.879	1.112
		HAS_RATING × *CHECKIN*	−2.198	1.472
		HAS_RATING × *LOCATION*	−0.473	1.156
		HAS_RATING × *VALUE*	2.059	1.165
		AFTER × *POOL*	6.789	4.455
		AFTER × *BEACH*	−9.047	10.31
		AFTER × *AC*	1.474	2.410
		Property image characteristics		
		log(*IMAGE_COUNT*)	4.530*	1.824
		BATHROOM_PHOTO_RATIO	3.431	8.526
		BEDROOM_PHOTO_RATIO	0.282	7.642
		KITCHEN_PHOTO_RATIO	15.57	11.28
		LIVINGROOM_PHOTO_RATIO	−10.39	8.484
		12 Human-interpretable image attributes		
Composition	1	*DIAGONAL_DOMINANCE*	2.516**	0.945
	2	*VISUAL_BALANCE_INTENSITY*	4.618***	1.350
	3	*VISUAL_BALANCE_COLOR*	8.869***	2.143
	4	*RULE_OF_THIRDS*	3.537**	1.106
Color	5	*WARM_HUE*	4.715*	2.363
	6	*SATURATION*	3.920*	1.927
	7	*BRIGHTNESS*	3.434*	1.679
	8	*CONTRAST_OF_BRIGHTNESS*	−4.897*	2.411
	9	*IMAGE_CLARITY*	6.212**	2.175
Figure-ground relationship	10	*SIZE_DIFFERENCE*	3.807*	1.541
	11	*COLOR_DIFFERENCE*	2.728*	1.372
	12	*TEXTURE_DIFFERENCE*	2.313*	1.090
Fixed effect			Property	
Seasonality			City-year-month	
Number of observations			76,901	
R^2			0.6670	

*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

accuracy of 90.4%, but it is not perfectly accurate;[12] future studies may further improve the classifier's performance with more training data. Second, we ignore an important element of the user search process on Airbnb. Typically, a potential guest browses several properties that appear on an Airbnb search page based on user-specified criteria (e.g., location, dates). A single image appears for each property on the search result page and may influence which properties the guest chooses to evaluate further (at which point the guest could see all images available for the property). Without access to consumer search processes, we cannot explicitly incorporate relevant information into our analysis, but future research may pursue such an analysis as more data (e.g., on the search process and transaction) become available to researchers. Lastly, as noted previously, our observational data do not enable us to make causal claims about the relationship between image features and property demand. Future research can investigate the causal impact of image attributes on property revenue if the data and context allow (e.g., in a randomized, controlled trial).

This paper relates property demand to both high-level and low-level dimensions of image features. Certain industries could benefit from the documented insights regarding the 12 image attributes. For example, home rental markets, such as Airbnb and VRBO, could reduce the issue of quality uncertainty by incentivizing their hosts to display high-quality property images. In the related industry of real estate (Zillow.com, Redfin.com, RE/MAX, etc.), a platform such as Zillow.com could use our results to improve listing images. Finally, our study is among the first to investigate the difference in demand generated by properties with verified and unverified photos and to examine the correlations between identifiable image attributes and property demand.

Endnotes

[1] See http://airhostsforum.com/t/professional-photography/3675/35.

[2] Of the 13,000 listings in our original data set, approximately 5,000 were already "treated" in January 2016. We exclude them from our main analyses, but as a robustness check, we repeat our main analysis with the pretreated units, and we obtain consistent results (see Section V.7 in the online appendix for details).

[3] We compute IVs based on property type, listing type, property capacity, and number of reviews, none of which are directly controlled by the host. Competitors are defined as the properties in the same zip code.

[4] The OpenEI data set provides average residential, commercial, and industrial electricity rates by zip code, compiled from ABB, Velocity Suite, and the U.S. Energy Information Administration data set 861: https://openei.org/doe-opendata/dataset/u-s-electric-utility-companies-and-rates-look-up-by-zipcode-feb-2011. Zillow Research provides average home values by zip code and home size (number of bedrooms): https://www.zillow.com/research/data/.

[5] The image data contain all images associated with all properties included in the data set. That is, they include images for properties that were verified before the observation started (and hence, are not included in the sample for the DiD analyses) and all images updated/added/deleted during observation periods.

[6] The vector *CONTROLS* includes two metrics that measure hosts' responsiveness, *RESPONSE_RATE* (percentage of messages/requests from guests that receive a response from the host) and *RESPONSE_TIME* (average number of minutes to respond to a guest). *MIN_STAYS* is the minimum number of nights that a guest can book. *MAX_GUESTS* is the maximum number of guests that may stay at once. *SECURITY_DEPOSIT* is the money that the guest will be charged if the host claims that the guest damaged the property and Airbnb approves the claim. *CANCELLATION_STRICT* is whether the cancellation policy is strict (one) or not strict (zero). *SUPER_HOST* is whether the host has a "super host" badge (one) or not (zero), which Airbnb assigns based on consumers' reviews, the host's responsiveness, etc.. *BUSINESS_READY* is whether the property has business-related amenities (one) or not (zero). *HAS_RATING* indicates whether the average guest ratings are presented on the property page (one) or not (zero).

[7] A possible explanation is that a change in photo quality does not necessarily reflect changes in the property or host. In a report on the smart pricing model, Airbnb did not describe property pictures as one of the many "factors at play" that the algorithm used: https://blog.atairbnb.com/smart-pricing/.

[8] The properties in our sample in the pretreatment period (January 2016) were priced at $179.5 per night on average.

[9] We do not include the review writing probability in the main DiD model as (1) review writing probability is indirectly controlled for through the number of reviews and property characteristics and (2) including it will significantly reduce the sample size. See Section V.9 in the online appendix for detailed discussions.

[10] For ease of understanding, we use standardized image attributes (i.e., variables are normalized to zero mean and unit variance).

[11] See the discussion at https://airhostsforum.com/t/airbnb-plus-program-anyone-else-who-thinks-the-new-photos-suck/30120.

[12] In our analyses, the *IMAGE_QUALITY* variable in the econometric model is the mean of the quality labels for the property's set of images. Although some images likely were misclassified during the machine learning step, some misclassifications should cancel out in the computed mean quality, leading to a relatively accurate mean value.

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